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R Shiny interactive web app

No web development skills required

SURVEY/MAPPING & PROPERTY DIVISION
Mapping & GIS Services
Section meeting presentation
Sarmen Bajari

Shiny is package that makes it easy to build interactive web apps straight from R & Python

Shiny is an R package that makes it easy to build interactive web

- Shiny is a free and open-source R package for developing web applications.
- It was announced by Joe Cheng, CTO of Posit, formerly RStudio, in 2012.
- In 2022, a separate implementation Shiny for Python was announced

Top Advantages of R Shiny framework:

- Inputs, outputs and display messages to make your data interactive on every device.
- Template layouts (Navbars, Sidebars, Tabs, Panels & Cards, Arrange Elements)
- Custom Shiny dashboards(using custom HTML, CSS, SCSS, JavaScript, and so on)
- Good documentation and big community.
- It looks like R Shiny apps loading faster than Esri web apps(it's just my opinion)

Hosting and deployment:

When it's time to put your Shiny app on the web, you can choose to deploy on your own servers or on Shiny hosting services(Shiny Servers , cloud, …)

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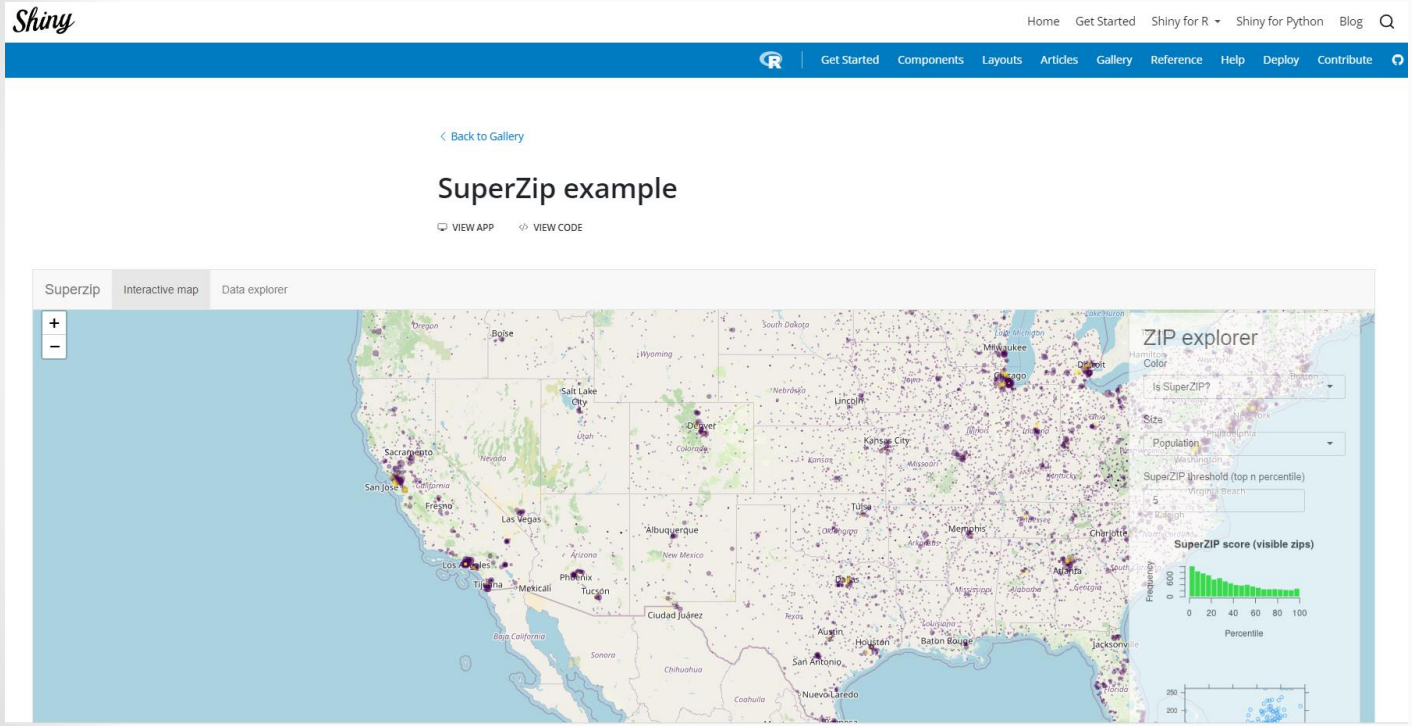
R Shiny Gallery Sample Apps with Code

You can easily copy/paste the raw code in RStudio

Link: Superzip (shinyapps.io)



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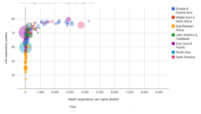
R Shiny Gallery Link:
https://shiny.posit.co/r/gallery/

Get Started | Components | Layouts | Articles | Gallery | Reference | Help | Deploy | Contribute

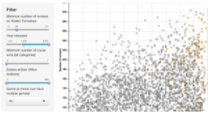
Interactive visualizations

Shiny is designed for fully interactive visualization, using JavaScript libraries like d3, Leaflet, and Google Charts.

Bus dashboard



Google Charts

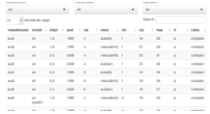


Movie explorer

SuperZip example

Widgets

Get to know many of the input and output widgets that are available in Shiny with these examples.



Basic DataTable

Custom input bindings

Custom input control

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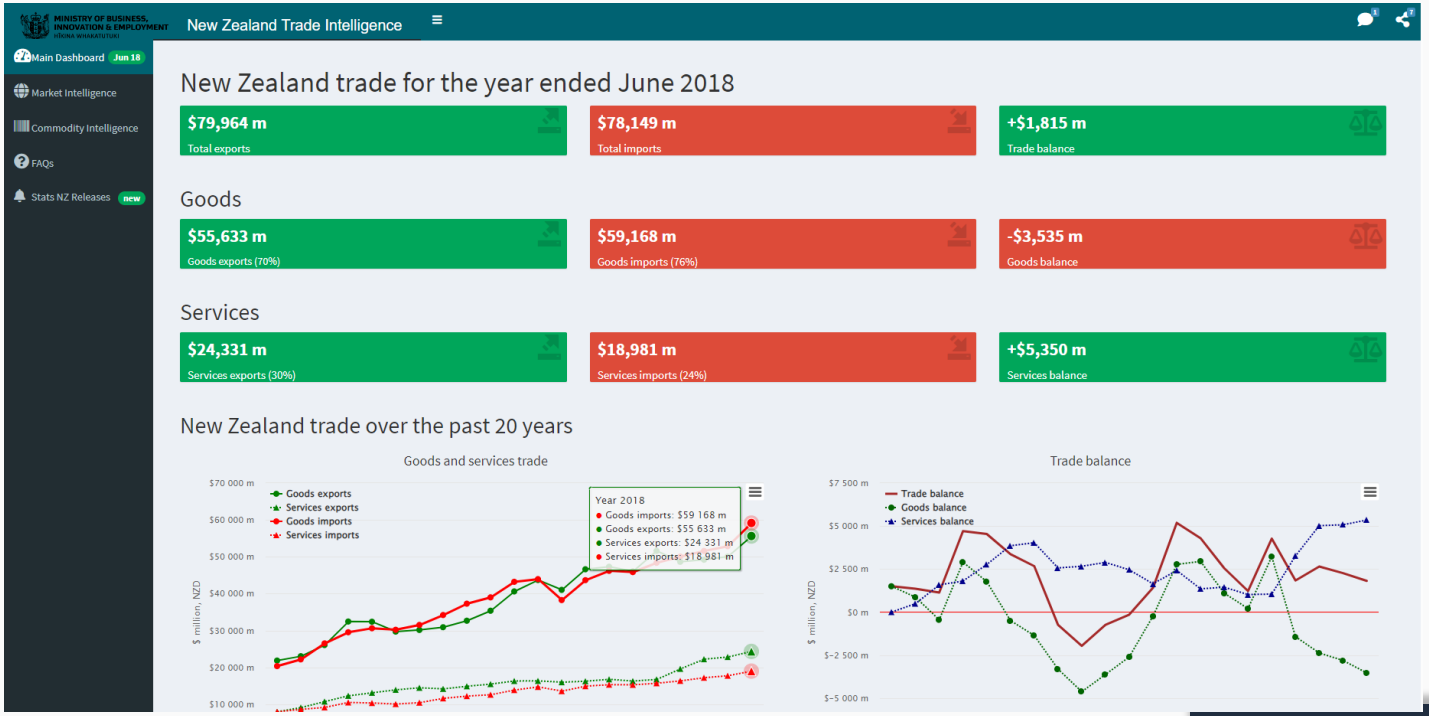
Sample Dashboard from New Zealand Government

Ministry of Business, Innovation & Employment



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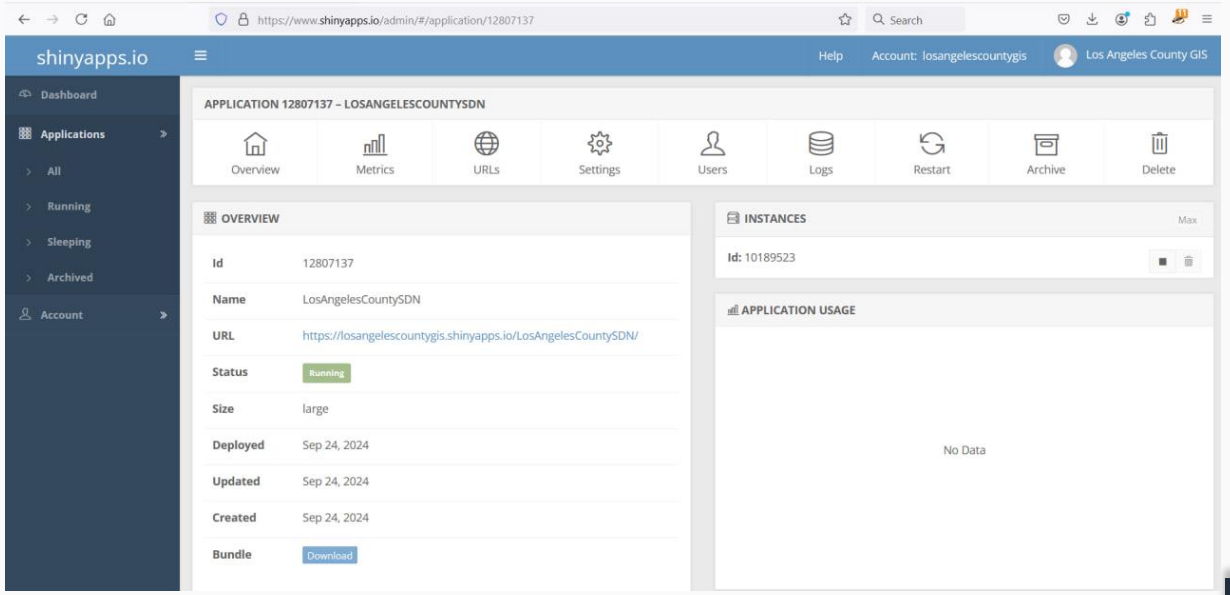
Los Angeles County Storm Drain Catch Basins Locations on an R Shiny app.

Link to test :
https://losangelescountygis.shinyapps.io/LosAngelesCountySDN/



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Los Angeles County Storm Drain Catch Basins Locations on an R Shiny app.

D:/RShiny - Shiny

http://127.0.0.1:4316 | Open in Browser | Publish

Los Angeles County Catch Basins

Show 10 entries

Search:

	OBJECTID	MAINTAINED	NAME	PERMIT_NO	DWGNO	PLAN_NO	LOCATED_IN	Disclaimer	Longitude	Latitude
1	1	CITY	LA City Catch Basin	Not Available		D-82	Los Angeles	Provided data is 'as is' without warranty of any kind	-118.31969007	33.98534472
2	2	CITY	LA City Catch Basin	Not Available			Los Angeles	Provided data is 'as is' without warranty of any kind	-118.58411641	34.15164567
3	3	CITY	LA City Catch Basin	Not Available			Los Angeles	Provided data is 'as is' without warranty of any kind	-118.28138442	33.96249254
4	4	LACFCD		1965421	PD031335		Carson	Provided data is 'as is' without warranty of any kind	-118.2635023	33.8239952
5	5	LACFCD		Not Available	PD031335		Carson	Provided data is 'as is' without warranty of any kind	-118.27603793	33.8220664
6	6	LACFCD		Not Available	PD031335		Carson	Provided data is 'as is' without warranty of any kind	-118.26811134	33.82389543

Search

65°F Mostly clear

ENG

12:03 AM 9/24/2024

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RStudio

RStudio IDE is an integrated development environment for R, a programming language for statistical computing and graphics.

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

app.R

```r
library(shiny)
library(readr)
library(leaflet)
library(DT)

# Define UI for the application
ui <- fluidPage(
  titlePanel("Los Angeles County Catch Basins"),
  leafletOutput("map"),
  DTOutput("table")  # Add output for the table here
)

# Define server logic required to create the map and table
server <- function(input, output) {
  # Read the data
  catch_basin_data <- read_csv("D:/RShiny/LosAngeles_County__StormDrain_CatchBasin.csv")

  # Check the structure of the data
  print(str(catch_basin_data))  # Inspect the data structure

  # Handle NA values in PERMIT_NO (optional)
  catch_basin_data$PERMIT_NO[is.na(catch_basin_data$PERMIT_NO)] <- "Not Available"
```

```r
install.packages('rsconnect')
library(rsconnect)
if(!require("devtools"))
install.packages("devtools")
devtools::install_github("rstudio/rsconnect")
install.packages("learnr")
load("D:/RShiny/.RData")
library(shiny); runApp('D:/RShiny/LosAngelesCountyGIS.R')
runApp('D:/RShiny/LosAngelesCountyGIS.R')
install.packages("DT")
library(shiny); runApp('D:/RShiny/LosAngelesCountyGIS.R')
runApp('D:/RShiny/LosAngelesCountyGIS.R')
runApp('D:/RShiny/LosAngelesCountyGIS.R')
```

```
R version 4.4.1 (2024-06-14 ucrt) -- "Race for Your Life"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

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  Natural language support but running in an English locale

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'citation()' on how to cite R or R packages in publications.
```

Name	Size	Modified
.RData	2.5 KB	Sep 24, 2024, 12:52 AM
.Rhistory	2.2 KB	Sep 24, 2024, 12:52 AM
ArcGIS		
Arduino		
Custom Office Templates		
desktop.ini	402 B	Oct 18, 2022, 10:30 PM
Visual Studio 2022		

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